21 November 1958
SUPERSEDING
NAVORD OS 5876B
1 August 1952

MILITARY SPECIFICATION

TEST SET MARK 177 MOD 0

1. SCOPE

1.1 This specification establishes the requirements for the procurement of Test Set Mark 177 Mod 0. This test set forms and checks the firing condensers used in A-5 and M-11 type firing mechanisms.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Fiber.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-131	Barrier Material; Water Vaporproof, Flexible.
MIL-V-173	Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.

FSC 6625

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MIL-S-6872	Soldering Processes, General Specification for.
MIL-P-17555(SHIPS)	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 275007	Test Set Mark 177 Mod 0.
LD 284866	Preparation for Delivery for Test Set Cable Assemblies.
1476194	Marking Instructions for Storage and Shipping Containers.

And all documents listed in the above LD's and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Test Set Mark 177 Mod 0 shall be manufactured in accordance with LD 275007 and all documents listed thereon.

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3.1.2 The requirements and procedures established in OP 400 shall be complied with as applicable.

3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.4 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings, shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.2 Moisture and fungus-resistant treatment. - Materials and parts of this test set which are not moisture and fungus resistant shall be treated, before assembly, with a varnish conforming to Military Specification MIL-V-173.

3.3 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.4 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, of drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.5 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Only non-corrosive and non-volatile fluxes shall be used. Rosin shall be used when soldering wires to contacts and terminals. The use of rosin core solder is permitted. Electrical connections shall be mechanically secure and electrically continuous before soldering. Sufficient heat shall be applied during the soldering process to avoid "cold" solder joints.

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3.6 Wiring. - All internal wiring shall be cabled wherever possible.

3.7 Performance requirements and product characteristics. The test set covered by this specification shall meet the following performance requirements and product characteristics.

3.7.1 Insulation resistance. - The insulation resistance between electrical circuits and between each circuit and the chassis shall be not less than 50 megohms when tested as specified in 4.5.3.

3.7.2 Power supply output. - The power supply output circuitry shall be properly connected as verified by testing as specified in 4.5.4.

3.7.3 V.T.V.M. calibration. - The test set meter shall indicate the same as the external voltmeter within ± 2 percent, when connected and tested as specified in 4.5.5.

3.7.4 S3 and S4 switch contacts. - External ohmmeter shall indicate 47,000 ±15 percent ohms and 1 megohm ± 5 percent, when connected and tested as specified in 4.5.6.

3.7.5 S5 switch contacts. - The S5 switch circuitry shall be properly connected as verified by testing as specified in 4.5.7.

3.8 Workmanship and cleanliness. - The test set shall be free from dust, dirt, metal filings, moisture, solder splashes, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIRMENTS.

4.1 Government inspector. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

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4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of one test set shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness, or participate in such inspections and tests on the sample as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor, prior to approval of the contracting activity or the completion of the tests on preproduction samples, shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Preproduction sample for a subsequent contract. The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

4.1.3 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

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4.3 Classification of tests. - The inspection and testint of mechanisms covered by this document shall be classified as follows:

(a) acceptance tests

(b) preproduction tests

4.3.1 Acceptance tests. - Acceptance tests for acceptance of the test set covered by this specification shall consist of tests to determine compliance with all requirements specified in this document.

4.3.2 Preproduction tests. - Preproduction tests are those performed on samples submitted in accordance with 4.1.2 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample.

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.1.1 Equipment required for performing these tests is as follows:

(a) One (1) megohmmeter-500-volts d-c. Range 10.0-100 megohms, accuracy within ±10 percent.

(b) One (1) d-c voltmeter. Range 0-150-volt range, accuracy within ±2 percent of full scale reading.

(c) One (1) ohmmeter. Ranges 0-100,000 ohms and 1-5 megohms, accuracy within ±3 percent of full scale reading.

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(d) 115-volt, 60-cycle, a-c supply.

4.5 Tests.

4.5.1 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ±10°F and ambient relative humidity.

4.5.2 Visual inspection. - The test set and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering and workmanship are in accordance with this document and the drawings.

4.5.3 Insulation resistance. - The test set shall be subjected to a conditioning period of 24 hours at a temperature of 77° ±2°F, and a relative humidity of 75 ± 5 percent. Using megohmmeter (item 4.4.1.1 (a)), the insulation resistance as specified in 3.7.1 shall not be less than 50 megohms.

4.5.4 Power supply output test.

(a) Set test set switches as follows:

1. ON-OFF at OFF.

2. CHARGE-CHECK at CHECK.

3. PLUG IN CONDENSERS-M-11 MECH. at M-11 MECH.

4. Selector at OFF.

5. HALF CHARGE-FULL CHARGE at FULL CHARGE.

(b) Connect CA-731 cable to test set AC PLUG and 115-volt power supply (item 4.4.1.1 (d)).

(c) Connect positive terminal of external voltmeter (item 4.4.1.1 (b)) to pin 6 of socket 1 on test set panel; connect negative terminal to pin 1.

(d) Set ON-OFF switch at ON. After warm-up period, voltmeter shall indicate 150 ± 7 volts.

(e) Set FULL CHARGE-HALF CHARGE switch at HALF CHARGE. Voltmeter shall indicate 75 volts ±5 percent.

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(f) Set HALF CHARGE-FULL CHARGE switch at FULL CHARGE.

(g) Remove positive terminal of external voltmeter from socket 1 and touch it to pin 6 of sockets 2 through 20. Voltmeter shall indicate 150 volts ±5 percent when terminal is touched to each socket.

(h) Remove external voltmeter.

4.5.5 V.T.V.M. calibration test.

(a) With selector switch at OFF, test set meter shall indicate zero. If meter does not indicate zero, it shall be possible to turn V.T.V.M. ADJ. control until it does.

(b) Connect positive terminal of external voltmeter (item 4.4.1.1 (b)) to pin 6 of socket 1; connect negative terminal to chassis.

(c) Set selector switch at CHARGING VOLTAGE. Test set meter shall indicate same voltage as external voltmeter. If meter does not indicate correctly, it shall be possible, after removing panel from case, to adjust METER ADJ. control until meter indicates correctly.

(d) Set ON-OFF switch at OFF and remove external voltmeter, and CA-731 cable.

4.5.6 S3 and S4 switch contacts test.

(a) Set PLUG IN CONDENSERS-M-11 MECH. switch at M-11 MECH.

(b) Set CHARGE-CHECK switch at CHARGE.

(c) Connect external ohmmeter (item 4.4.1.1 (c)) terminals to pins 3 and 6 of socket 1. Meter shall indicate 47,000 ohms ±15 percent.

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(d) Remove ohmmeter terminals from pins 3 and 6 and connect to pins 5 and 6. Meter shall indicate 47,000 ohms ±15 percent.

(e) Repeat steps (c) and (d) for sockets 2 through 20. Meter shall indicate 47,000 ohms ±15 percent when connected to each pair of pins.

(f) Set CHARGE-CHECK switch at CHECK.

(g) Set PLUG IN CONDENSERS-M-11 MECH. switch at PLUG IN CONDENSERS.

(h) Connect external ohmmeter terminals to pins 3 and 6 of socket 1. Meter shall indicate 1 megohm ±5 percent.

(i) Remove ohmmeter.

(j) Repeat steps (h) and (i) for sockets 2 through 20. Meter shall indicate 1 megohm ±5 percent when connected to each socket.

4.5.7 S5 switch contacts test.

(a) Connect CA-731 cable to test set AC PLUG and 115-volt a-c supply (item 4.4.1.1 (d)).

(b) Set ON-OFF switch at ON.

(c) Set CHARGE-CHECK switch at CHARGE.

(d) Set selector switch at OFF.

(e) Turn V.T.V.M. ADJ. control until meter indicates zero.

(f) Set selector switch at CHARGING VOLTAGE. Test set meter shall indicate 150 volts ±5 percent.

(g) Repeat step (f) with selector switch set at positions 1 through 20. Test set meter shall indicate 150 ±7 volts at each switch position.

(h) Set CHARGE-CHECK switch at CHECK.

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(i) Set PLUG IN CONDENSERS-M-11 MECH. switch at M-11 MECH.

(j) Set selector switch at each of its numbered (1 through 20) positions. Test set meter shall indicate zero at each switch position when shorting pins 3 and 1 of sockets 1 through 20 in turn (corresponding to selector switch positions).

(k) Set selector switch and ON-OFF switch at OFF.

(l) Remove CA-731 cable.

4.5.8 Acceptance criteria. - Failure of any test set to meet any of the requirements and tests of this specification shall result in rejection of the test set. If the number of failing test sets exceeds the number permitted by the associated criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, drying, and preservative application. - (Not applicable)

5.1.1.2 Unit packaging. - Prior to unit packaging, all test set cables shall be packaged in accordance with Bureau of Ordnance LD No. 284866. Each test set shall be packaged in a Type I, Class 1, Style RSC corrugated fiberboard box, having a minimum Mullen test of 275 pounds, conforming to the requirements of Federal Specification PPP-B-636. Sufficient fillers and folded corner pads, fabricated from the same box material, shall be utilized within the carton to provide a tight pack and to prevent direct contact between protrusions of the test set case and the walls of the fiberboard box.

5.1.1.2.1 The unit package, prepared as described above, shall be further packaged Method IA-14 conforming to the requirements of Military Specification MIL-P-116. The inner carton with all corners blunted shall be inserted into a water-vaporproof bag fabricated from Class I barrier material conforming to the requirements of Military Specification MIL-B-131, and closure heat-sealed. The outer carton shall be a Class 2, Style RSC fiberboard box conforming to the requirements of Federal Specification PPP-B-636, W5c or W5s compliance symbol.

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5.1.1.3 Intermediate package. - (Not applicable)

5.1.2 Level B. - (Not applicable)

5.1.3 Level C.

5.1.3.1 Cleaning, drying, and preservative application. - (Not applicable)

5.1.3.2 Unit packaging. - Unit packaging shall be in accordance with Method III of Military Specification MIL-P-116. The unit package shall contain one (1) test set.

5.1.3.3 Intermediate package. - (Not applicable)

5.2 Packing.

5.2.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Military Specification MIL-P-17555.

5.2.1.1 Exterior containers. - Each test set, individually packaged as specified above for Level A protection, shall be packed in an overseas type wood-cleated plywood box or in a Class 2, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

5.2.1.1.1 When a nailed wood box is selected to be used for shipment, box lumber shall conform to Group I, II or III woods of Federal Specification PPP-B-621. The lumber shall be of one (1) inch nominal thickness and surfaced on both sides.

5.2.2 Level B. - In addition to the following, Level B packaging shall be in accordance with Military Specification MIL-P-17555.

5.2.2.1 Exterior containers. - Each test set, individually packaged as specified above for Level A protection, shall be packed in a domestic type wood-cleated plywood box or in a Class 1, Style 2, 2 1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specifications PPP-B-601 or PPP-B-621, respectively.

5.2.3 Level C.

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5.2.3.1 Exterior containers. - Each test set, packaged in accordance with 5.1.3.2, shall be packed in a manner which will provide protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special marking. - The following precautionary marking shall be stenciled or stamped on the top, one (1) side and one (1) end of the shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

5.3.2 Normal marking. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129, including serial numbers.

5.3.2.1 For detailed marking requirements, see Bureau of Ordnance Drawing Number 1476194.

6. NOTES

6.1 Procurement documents shall specify the title, number, and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents shall also specify the desired level of packaging and packing. The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116, and to paragraph 6.1.1 of this document.

6.1.1 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

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For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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